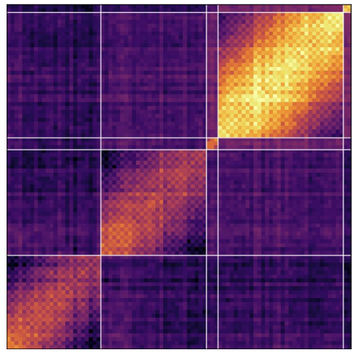
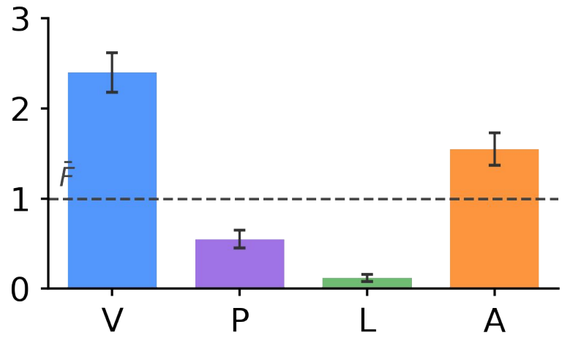
Tier-1: Reference-based Target vs. cached source features



$1 - \mathrm{CKA}(r)$

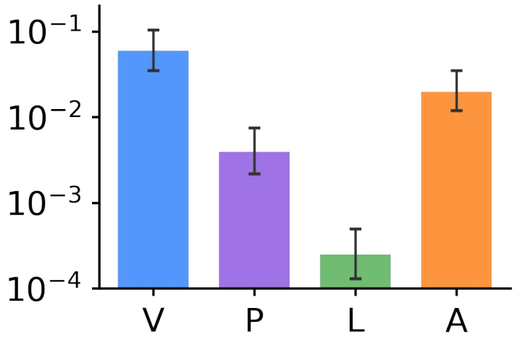
Representation shift



$F(r)$

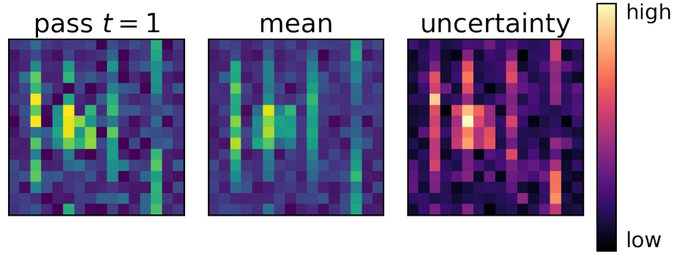
Parameter sensitivity

Tier-2: Reference-free From the target data alone



$G(r)$

Gradient norm

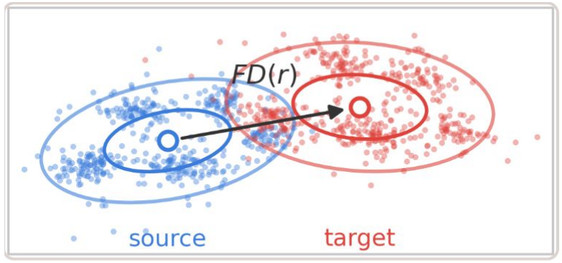


$\hat{U}(r)$

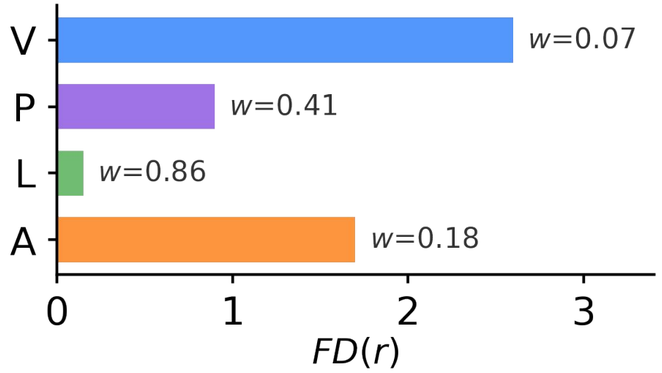
MC-dropout uncertainty

Fréchet-distance gate, $w(r)$

Smaller source–target gap → trust Tier-1 more



Domain gap $FD(r)$



Per-region adaptation cost, $D(r)$ — gated blend of the two tiers